

- Product Requirements Document

Livelihood Installation App

Document Version	v1
Date	July 2026
Product Area	Livelihood Digitisation Programme
Platform (Phase 1)	Android App (Field Technicians); Web App (Project Manager, Installation Reviewer)
Status	For Review
Preceded By	Setu4Livelihoods (Livelihood Issue Resolution Platform)
Prototype link	https://installation-module-livlihood-platform-app-1068851155482.asia-southeast1.run.app

1. Executive Summary

The Livelihood Installation App is the operational tool SELCO teams use to plan, execute, and record installations of solar systems and livelihood machines for end users under the Livelihood Programme. It is the second digital tool built for the programme, after the Setu4Livelihoods issue resolution platform.

Project Managers use the app to plan work and dispatch it to vendors. Field Technicians use it on their phones to complete installation reports on site, including when they have no internet. An Installation Reviewer approves completed reports, hands the record over to the Asset Management module, and marks that specific asset as ready for O&M on Setu4Livelihoods and in the WhatsApp chatbot — where the end user can then raise maintenance tickets and track resolution over the life of the asset.

2. Problem Statement

Livelihood installations today are tracked through paper forms, shared spreadsheets, and phone calls. Programme teams live with the following consequences:

- There is no single digital record of whether an installation has been completed for a given end user or asset.
- Assigning solutions and vendors across dozens of end users is done in offline spreadsheets, so the planning data never enters a system of record.
- A Project Manager has no easy way to find the right solution for each end user based on sector, value chain, and sunshine hours.

3. Goals & Success Metrics

3.1 Product Goals

- Record every installation digitally, from project setup through final approval.
- Let a single Project Manager plan a 50–100+ end user Installation Plan (solutions, vendors, templates) without leaving the app.
- Let Field Technicians work fully offline and sync when they are back online.
- Ensure each role sees only what it needs to do its job.
- Enable O&M support the moment an installation is approved.

3.3 Proposed Workflow

The diagram below shows the flow the app introduces — how a Project breaks down into an Installation Plan, then into End User Sites, each assigned a Solution, and how each Solution further expands into [UPDATED] one or more Machines and a Solar bundle that get assigned to Vendors. Today, this flow is done manually across paper forms, spreadsheets, and phone calls; the app makes it a single connected workflow inside one system.

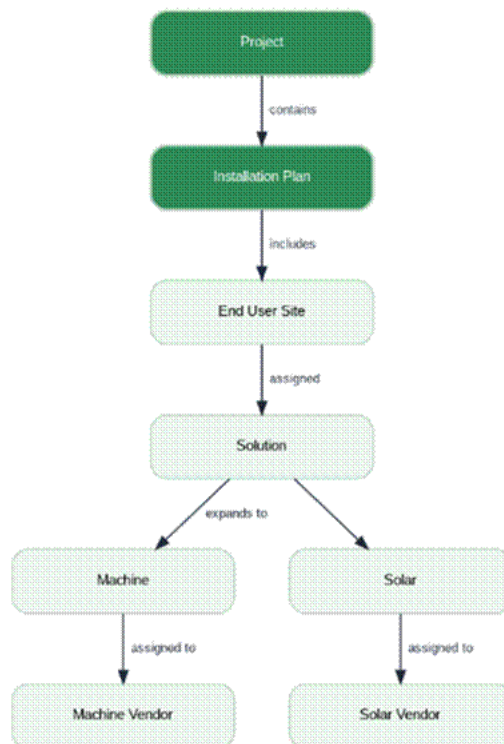


Figure 1. Proposed workflow. Each Project holds one or more Installation Plans. A Plan includes multiple End User Sites, each of which is assigned a Solution. Every Solution expands into multiple assets — Machines and a Solar bundle — which are then assigned to Vendors (the same Vendor can cover both, or different Vendors for each).

4. Scope

4.1 In Scope (Phase 1)

- Master data setup: sector-to-solution mapping and Project ID generation.
- Project creation by the Project Manager, including selecting end user sites into scope via Excel.
- Installation Plan creation, including solution assignment (Sheet 1), vendor assignment (Sheet 2), installation template configuration per unique solution, and publishing.
- Initial solution ingestion limited to 14 widely-used solutions (per Scale team) for testing, ahead of full repository ingestion.
- Validation preventing an end user site from being in two active Installation Plans at once.
- Android app for Field Technicians to complete IC Reports.
- Web app for Project Managers and Installation Reviewers.
- Full offline capture on the Android app, with manual sync when internet returns.
- Two IC Report formats — Solar Installation Report and Machine Installation Report (detailed formats in separate documents).
- [NEW] The IC Report automatically shows the right parts list and fields for whichever Solution is assigned to that end user (see FR-01a, FR-11a and Section 13).
- Handover Letter generation from image as part of the IC Report.
- Installation Reviewer role with its own login (approve or return with reason).
- Automated notifications for key lifecycle events (see Section 9).
- Handoff of approved installations to the Asset Management module.
- End user site data ingestion from the Admin module.

4.2 Out of Scope (Deferred)

- Web app for Field Technicians (Phase 2).
- Native iOS application (Phase 3).
- Background auto-sync for offline reports (manual sync only in Phase 1).
- GPS tagging and real-time installation progress tracking.
- Analytics dashboards for programme managers.
- Modifications to already-approved installations.
- Field-level lock/edit rules on installation templates during installation.
- Governance of the Solution Repository (who maintains it and how it is updated).
- Detailed field-by-field format of the Solar and Machine IC Reports (maintained separately).

5. User Roles

The app has three roles. Each has its own login.

Role	Who is this?	What they can do	Where they work
Project Manager	SELCO staff who plans installations.	Create Projects and Installation Plans; assign solutions and vendors;	Web app.

		configure installation templates; publish plans.	
Field Technician	On-ground staff (usually employed by a Vendor Organisation) who does the installation.	See assigned tasks; fill and submit IC Reports; work offline and sync later.	Android app.
Installation Reviewer	SELCO staff who checks and approves submitted reports.	Review submitted IC Reports; approve, or return with a reason.	Web app.

Vendor Organisation is a data entity, not a role. Vendors are assigned to installation tasks via Vendor Assignment (FR-07) and receive notifications, but do not log into the app directly. The people who log in on behalf of a Vendor Organisation are Field Technicians.

6. Core Data Entities

These are the main records the app creates and manages.

Entity	Description	Key Attributes
End User Site	The physical location where an installation happens, linked to an end user.	Name, location (state/district/block), sector, value chain, sunshine hours (auto-set from state), current lock status.
Project	The top-level record grouping all installations under one programme intervention.	Name, justification code, start/end dates, geography (one or more states/districts/blocks), auto-generated Project ID, Project Manager, end user sites in scope.
Installation Plan	A plan under a Project that covers one or more end user sites for installation.	Linked Project, plan geography (subset of Project geography), sector(s), assigned Installation Reviewer, planned start/end dates, senior programme contact for escalations, publish status.
Solution	[UPDATED] A ready-made package of one or more machines plus an associated solar bundle, for a specific livelihood use case (most Solutions have one machine; some, like Multi-Stage Processing, have more - see FR-01a).	Sector, value chain, sunshine-hours range, machine specs, solar bundle component list.
Machine / Asset	A single asset expanded from a Solution during Vendor Assignment. Each Solution	Type (machine name or 'Solar'), parent Solution, assigned Vendor, installation status.

	expands into a Machine row and a Solar row per end user site.	
Vendor Organisation	The company that installs assigned equipment.	Name, contact list (multiple emails), assigned states, asset-type eligibility.
Installation Template	The data ingestion form the Project Manager fills for each unique Solution in an Installation Plan; pre-populates IC Report fields for the technician.	Bound to a Solution, contains a machine section and a solar section, filled per Plan (blank each time).
IC Report	The Installation Completion Report filled by the Field Technician per assigned task.	Format (solar or machine), pre-filled template data, captured data, photos/video, reviewer status.
Handover Letter	A formal document generated when installation is complete and approved.	End user details, asset details, date, signatures.

7. Functional Requirements

7.1 Master Data & Configuration

These configurations must be in place before any Project or Installation Plan can be created.

FR-01: Sector-to-Solution Mapping & Solution Filtering

The Solution Repository holds many solutions across four sectors, and a Project Manager cannot realistically scroll through all of them to find the right one for each end user site. This FR narrows the list automatically so that when a Project Manager assigns a Solution in the Installation Scope Excel (FR-05), the dropdown for each row shows only Solutions that fit that specific site.

The narrowing runs per end user site — not once for the whole Installation Plan — because a single Plan can span multiple states and multiple sectors. A site in Karnataka under Agriculture and a site in Assam under Textile & Crafts, sitting in the same Plan, each need their own filtered list.

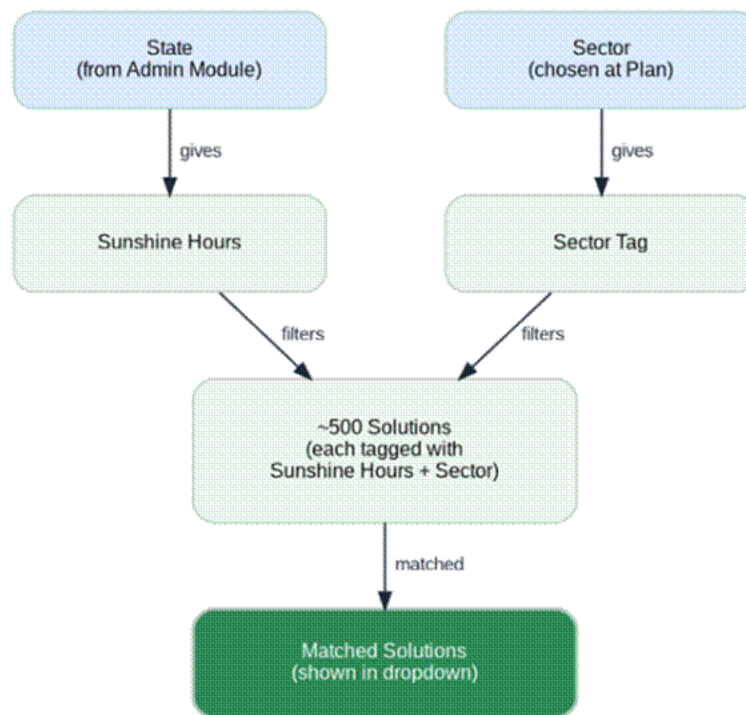


Figure 2. State (from the Admin Module) contributes Sunshine Hours for each site; Sector (chosen at Installation Plan creation) contributes the Sector Tag. Both filter the master pool of ~500 tagged solutions, and only the matching ones appear in the Project Manager's dropdown.

Narrowing criteria, in this order:

- Sunshine Hours — set automatically for each site during ingestion (FR-15), based on the site's state.
- Sector — every Solution is tagged with a sector; only Solutions matching the site's sector remain.

Phase 1 solution set: For initial rollout, solution ingestion is limited to the 14 solutions already in wide use by the Scale team, rather than the full ~500-solution repository. These 14 are ingested first and used for testing; the remaining solutions will be added to the repository in upcoming phases.

[NEW] FR-01a: Automatically Splitting Machine and Solar Parts from the Ingestion Sheet

Every Solution's ingestion sheet (Section 13) is split into a Solar Parts List and an Associated Machines list. The system reads this automatically when the Solution is added and tags each item Solar or Machine - no manual tagging, and it's done once per Solution, not once per Installation Plan.

1. Solar Parts List items are tagged Solar and go into that Solution's Solar installation.
2. Each item under Associated Machines is tagged Machine and becomes its own Machine installation. Most Solutions have one machine (1 Machine + 1 Solar = 2 installations); some, like Multi-Stage Processing, have more than one (2 Machines + 1 Solar = 3 installations).

3. This tag drives everything downstream: Vendor Assignment rows (FR-07), Installation Template sections (FR-08), and what each Field Technician sees in the IC Report (FR-11a).

FR-02: Automatic Project ID Generation

Every Project needs a unique identifier that programme teams can reference in conversations, reports, and files. The system generates this ID automatically on Project creation, and once generated it is read-only — no one can edit it, which protects the integrity of downstream references.

The ID encodes structured information about the Project (programme prefix, justification, financial year) so that even without opening the record, a reader can decode what kind of Project it is. Because a single Project can now span Installation Plans in different sectors (FR-04), Sector is no longer part of the Project ID — it lives at Plan level instead.

Rules:

- Format: PROJ-<Justification Code>-<Financial Year>-<Sequence Number>
- Example: PROJ-SD-26-27-001
- The Sequence Number auto-increments and resets per unique combination of Justification Code and Financial Year.
- Once generated, the Project ID is read-only.

7.2 Project Creation

FR-03: Project Creation

A Project is the top-level container for one programme intervention — it groups everything that happens for a given funding cycle, geography, and set of end users. Creating a Project happens in three sub-steps: entering basic details, defining geography, and selecting which end user sites are in scope. Only after all three are done is the Project ID generated and the Project ready to hold Installation Plans.

End user site selection is done through an Excel round-trip because a Project may cover hundreds of sites and marking them one by one in a UI would be tedious. The Excel is pre-populated with every site available in the selected geography, and the Project Manager only has to mark Include (Yes/No).

Sub-step 1 — Project Details:

- Project Name, Justification Code, planned Start Date and End Date.

Sub-step 2 — Geography:

- Select one or more States, Districts, and Blocks. All three levels support multi-select.

Sub-step 3 — Project Scope:

- Download the End User Site Excel. It lists every site in the selected geography, pulled from the Admin module (FR-15).
- Mark each site with Include (Yes/No) and upload the completed Excel.

- If any row fails validation (e.g. edited read-only columns), the whole upload is rejected and a per-row error report is shown on the same screen. The PM fixes the Excel and re-uploads.
- Included sites become the pool from which Installation Plans can draw.

Notes:

- On successful upload, the Project ID is generated (FR-02).
- A single Project can hold multiple Installation Plans across different sectors — Sector is not captured at Project level.
- Start Date auto-fills to today's date when the form opens. End Date is entered manually by the Project Manager — it is not auto-filled.
- All fields in Project creation (Sub-steps 1–3) are mandatory unless stated otherwise.

7.3 Installation Plan Creation

Under a Project, the Installation Plan is the actual unit of work dispatched to vendors. Building a Plan is a five-step sequence: setup, solution assignment, vendor assignment, template configuration, and publish. Each step blocks the next — no vendors can be assigned until solutions are set, no templates can be filled until the unique solutions in the Plan are known, and the Plan is not visible to vendors until it is Published.

FR-04: Installation Plan Setup

Setup defines the boundary and ownership of the Plan — where it will operate, which sectors it covers, who will review the resulting reports, and who to escalate to if the Plan runs late. This is the first thing a Project Manager does when creating a Plan, and the details entered here shape everything that follows.

Fields to enter:

- Plan geography - States, Districts, and Blocks. Must be a subset of the parent Project's geography.
- One or more Sectors. A plan can span multiple sectors, just as it can span multiple states.
- Assigned Installation Reviewer — the specific person who will review IC Reports submitted under this Plan.
- Planned Start Date and End Date.

Notes:

- Start Date auto-fills to today's date when the form opens. End Date auto-fills to a date one month from today. Both dates remain editable before the Project Manager submits.
- All fields in Installation Plan Setup are mandatory.

FR-05: Installation Scope & Solution Assignment (Sheet 1)

Once a Plan is set up, the Project Manager needs to pick which end user sites (from the Project's scope) are in this Plan and assign a Solution to each site. Doing this one by one in a UI does not scale — Plans routinely include 50–100+ sites, and Solution assignment is highly repetitive (many

sites often get the same Solution). An Excel round-trip lets the Project Manager use drag-fill and copy-paste to move quickly across similar sites.

The Excel is pre-populated with every site available in the parent Project. Two columns matter: Include (Yes/No) and Solution. These are linked — a Solution can only be assigned to a site that is being included in the Plan. To enforce this cleanly, the Solution column is conditionally available: it opens only when Include = Yes, and it stays locked when Include = No.

This prevents the Project Manager from accidentally assigning a Solution to a site that will not actually be installed, and keeps the Excel visually clear about which rows are in scope. The Solution dropdown itself is also filtered per row per FR-01 so the Project Manager can never accidentally assign an incompatible solution.

The flow:

- Download the Installation Scope Excel. Read-only columns: Site Name, Village, State, District, Block, Sector.
- Editable columns: Include (Yes/No, defaults to No on download); Solution (dropdown, filtered per row per FR-01, disabled when Include = No).
- Change Include to Yes for the sites to be included. As soon as Include is set to Yes for a row, the Solution cell for that row becomes active and the dropdown can be opened.
- Pick a Solution for each Include = Yes row. Excel drag-fill and copy-paste can be used across similar rows.
- Upload. The system runs the validations below; if all pass, Solution choices are locked for the Plan and the Vendor Assignment Excel (FR-07) is generated.

Validation rules (all enforced on the same upload screen):

- Every Include = Yes row must have a Solution assigned. Blank Solution on an included row is a validation error.
- No Include = No row may carry a Solution value. If a Solution has been entered on a Not-Included row (through paste, direct edit, or leftover data), it is a validation error.
- Every Solution value on an Include = Yes row must be a valid Solution per FR-01 for that site.
- Every Include = Yes site must pass the end user site lock check (FR-06).

If any validation fails, the upload is rejected in full — no partial acceptance. The system returns a per-row error report on the same upload screen, listing every failing row with its site name, the failure reason, and what needs to change. The Project Manager fixes the Excel and re-uploads.

Post-Publish Scope Edits:

Once an Installation Plan has been Published (FR-09), its scope cannot be freely re-opened — a vendor has already been dispatched based on the published site list, and letting the Project Manager add sites back in through the same casual Excel round-trip would undermine that commitment. Removing a site that is no longer going ahead is a different, lower-risk action than adding one, so the two are treated asymmetrically. Both are still done through the same Installation Scope Excel (FR-05), re-downloaded and re-uploaded for this purpose.

- Include: Yes → No (removing/cancelling an end user site from the Plan) is allowed at any time after Publish.

- Include: No → Yes (adding a new end user site to an already-Published Plan) is not allowed. A Project Manager who needs to add a new site must do so through a new Installation Plan.
- Validation: if an IC Report has been submitted for the end user site — for either the Machine or the Solar asset — that site can no longer be removed. Once field execution has been reported for either asset, Include is locked and a Yes → No edit on that row is rejected as a validation error.
- This is the same Excel step where Solution is assigned (Sheet 1); Solution values are not editable after Publish, only the Include column.

FR-06: End User Site Lock Validation

An end user site that is already being installed under one Installation Plan cannot be pulled into another Plan at the same time. Without this rule, two Project Managers could unknowingly dispatch different vendors to the same end user, which would confuse the vendors, the technicians, and the end user themselves. The lock is a hard block — there is no override — because the operational cost of a double-booking is high enough that even edge-case overrides would erode trust in the system.

The lock applies only at the Installation Plan level. The same site can still appear in the scope of multiple Projects, because Project scope is a much broader concept than an active installation.

Rules:

- The lock is checked during Installation Scope Excel upload (FR-05).
- A site is locked if any of its installation reports is in one of these statuses: Pending, Submitted, Under Review, or Changes Requested.
- It becomes available again once every installation report for that site has been Approved.
- The lock is a hard block — no override option is provided.

When a locked site is caught during upload, the system returns a per-row report showing:

- Site Name; Project Name (the other Project holding it); Installation Plan (the other Plan); Current Status; and Validation Message.
- Sample message: End User Site "ABC Farmer Group" is currently undergoing installation under Installation Plan "IP-2026-001". The site can be included in another Installation Plan only after all installation reports are approved by the assigned Installation Reviewer.

FR-07: Vendor Assignment (Sheet 2)

[UPDATED] Once Solution assignment is accepted, the system expands each Solution into one row per Machine plus one Solar row, per end user site (see FR-01a) - most Solutions produce 2 rows, Solutions with more than one machine produce more. The system does this automatically - the Project Manager doesn't build these rows by hand. Vendor assignment is done per asset row,

The flow:

- Vendor assignment takes place in the UI immediately after the solution ingestion template is uploaded. Once the solution is expanded (e.g., Solar, Machine), the user can assign the installation vendor. Upon selecting a vendor, the Vendor Organization field (dropdown listing

all available vendors) and the Vendor Email ID field (dropdown listing the contacts associated with the selected vendor) will be displayed.

Setu

KA-2626-379-00099
New Installation Plan

Plan Details Installation Scope Template **Technician Assignment**

End user	Solution	Vendor Organization	Email ID
Amit Singh +91 887963544	Solar Pump	Select Vendor	email@example.com
	Solar Home Light	Select Vendor	email@example.com
	Fan	Select Vendor	email@example.com
	Solar Pump	Select Vendor	email@example.com

< Back Submit >

FR-08: Installation Template Configuration

A key goal of this app is reducing what the field technician has to type on-site. To do that, the Project Manager pre-fills a data template for each Solution used in the Plan, spelling out the components, quantities, makes, models, and specifications that the technician will install. The technician's IC Report is then pre-populated with this data — they just confirm or correct each field based on what actually got installed, rather than typing everything from scratch.

Templates are per unique Solution, not per end user site. If 40 sites in the Plan all use the Pulverizer Solution, the Project Manager fills the Pulverizer template once and it applies to all 40. [UPDATED] Each template has one section per Machine plus one Solar section (see FR-01a) - most Solutions need two sections, Solutions with more than one machine need more. Templates start blank on each new Installation Plan — there is no versioning or carry-over from earlier Plans, which keeps the model simple and avoids stale data risks.

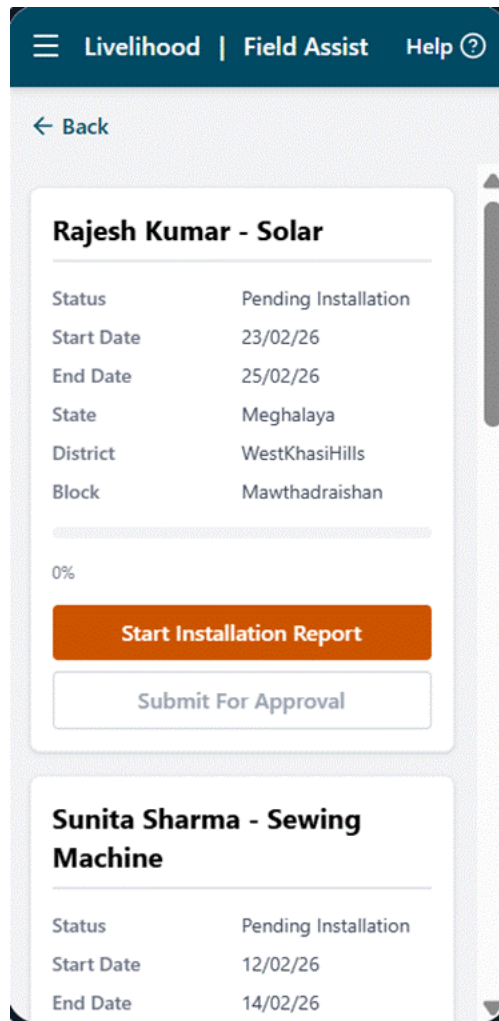
The flow:

- After Solution assignment is accepted (FR-05), the system identifies all unique Solutions in the Plan. If the Plan uses Pulverizer, Grinder, and Chaff Cutter, that is 3 unique Solutions and 3 templates.
- The Project Manager downloads one blank pre-populated template per unique Solution.
- Fill each template with: Installation Components, Quantity, Make, Model, Capacity, Technical Specifications, and other installation parameters.
- Upload each completed template. If a template fails validation (e.g. required fields missing), that upload is rejected in full and a per-cell error report is shown on the same screen. The PM fixes the template and re-uploads.

[NEW] Figure 7. Installation Plan creation - Template step (step 3 of 4). The Project Manager sees one card per Solution in the Plan, each with its own Download Template / Upload Template controls - this is the actual screen behind the download-fill-upload flow described above.

How the template is used by technicians:

- On upload, the template's machine section is bound to the machine vendor's task; the solar section is bound to the solar vendor's task.
- Each technician sees only the section for their assigned scope. A machine vendor's technician sees the machine section; a solar vendor's technician sees the solar section.
- Any edits the technician makes stay on their IC Report — they do not back-propagate to the template.
- Field-level lock/edit rules on the template during installation are out of scope for this PRD.
- **[NEW] Because each Solution's parts are pre-tagged Machine or Solar (FR-01a), the template - and the IC Report built from it (FR-11a) - automatically shows only the sections and parts that Solution needs, with no manual setup by the Project Manager.**
- **[NEW] In the Field Assist app, every task card clearly shows the end user's name and what is being installed - for example “Rajesh Kumar – Solar” or “Sunita Sharma – Sewing Machine”. This way the Field Technician always knows which end user and which asset (Machine or Solar) a task is for, before even opening it. See Figure 6 and FR-10.**



[NEW] Figure 6. Field Assist app task list - each card shows the end user's name and what is being installed (Solar, or the machine name).

FR-09: Publish Installation Plan

Publishing is the moment an Installation Plan becomes live — vendors get their tasks and field work can begin. Before that moment, the Plan is a draft that only the Project Manager can see. Making Publish an explicit step (rather than implicitly firing on Vendor Assignment upload) gives the Project Manager one final chance to review everything, and gives the system one clear moment to run a full validation and dispatch tasks atomically.

Publish flow:

- The system runs a Publish validation: every included site has a Solution, every expanded row has a Vendor and Vendor Email, and every unique Solution has a filled template.
- When the Project Manager clicks Submit at this final step, the system shows an Important Note confirmation modal before publishing: “Once this Installation Plan is submitted, you won’t be able to add any new end-user sites. You can still remove an existing site, but only if no Installation Report has been submitted for it.” with Cancel and Confirm & Submit options.
- On Confirm & Submit, the system publishes the Plan.

- On Publish, installation tasks are automatically assigned to the respective Vendors and appear in their Field Technicians' task lists (FR-10).
- Vendors receive notifications for their assigned tasks (see Section 9).

7.4 Installation Execution (Field Technician)

FR-10: Task Assignment & Visibility

A Field Technician's Android app is their inbox of assigned installation work. They see only the tasks assigned to their Vendor Organisation, and each task carries all the context and pre-filled data they need to install without going back to the Project Manager for clarifications.

What the technician sees on each task:

- End User Site name and location; the type of asset to install (Machine name or Solar); the Solution name.
- The template data prepared by the Project Manager (FR-08), pre-filled and ready for confirmation.
- Only the section (machine or solar) relevant to their assignment. Machine vendors do not see the solar scope; solar vendors do not see the machine scope.
- Site details pre-filled from ingestion — no re-entry needed.

FR-11: Installation Completion Report (IC Report)

The IC Report is the technician's record of what they installed. It is filled on-site in the Android app and eventually becomes the source of truth for that installation in the Asset Management module. Submission works in two ways depending on connectivity at the site: if the technician has internet, they tap Submit and the report goes directly to the Reviewer's queue. If they don't, the report saves on the phone and the technician taps Sync later, once they're back in a connected area, to push it up.

There are two report formats to cover the two kinds of assets: a Solar Installation Report and a Machine Installation Report. The right one is selected automatically based on the Asset Type of the task. The detailed field-by-field format of each report is maintained as a separate document, outside this PRD.

[NEW] See Section 13.6 for a summary of the actual Solar IC Report and Section 13.7 for the actual Machine IC Report - both linked in full in Section 13.5.

[NEW] FR-11a: The IC Report Changes Automatically Based on the Solution

The IC Report automatically matches whichever Solution is assigned to the end user - showing only what that Solution needs:

1. Only the parts, categories, product names, sizes, and quantities that apply to that Solution are shown - the Field Technician only sees and fills what's relevant to their Solution.
2. Anything that doesn't apply to that Solution is left out of the report entirely.

Note: this data comes from the Solution Ingestion Sheet (Section 13); the sheet only supplies the data - the IC Report's actual layout is defined separately (FR-11).

[NEW] FR-11b: Dynamic Image Capture Checklist Generation

Just like the rest of the IC Report (FR-01a, FR-11a), the “Images to be Captured During Installation Visit” section is built automatically from whichever components are actually part of the selected Solution, using an Image Capture Master that maps each required photo to a Solution component:

- The IC Report only includes photo entries whose mapped component is present in the selected Solution; for each one, the heading, instructions, and default number of photos are filled in automatically. Entries for components not in that Solution are left out entirely - the same logic as FR-01a and FR-11a.
- Two photos are always required, for every Solution, no matter what components it has: a clear image of the End User place (Long Shot), and a clear image of the End User with Solar System.
- If the Image Capture Master is updated later, that change applies automatically to the next report generated - no changes to the report template itself are needed.

Note: this is the mechanism behind “Required photos” in Section 13.6/13.7 - the actual reference table is in Section 13.8.

How the report works:

- The IC Report is pre-populated with the template data prepared by the Project Manager (FR-08). The technician confirms or edits each field based on what was actually installed on-site.
- Photos and (where applicable) video are uploaded as part of the report.
- A Handover Letter is generated as part of the IC Report on submission.
- [NEW] The IC Report combines pre-filled data (FR-08) with whatever the technician adds on-site, including scanned serial numbers for the Annexure (see Section 13.5) - the final report only exists once submitted.

[UPDATED] Installation Verification Code (End User Confirmation):

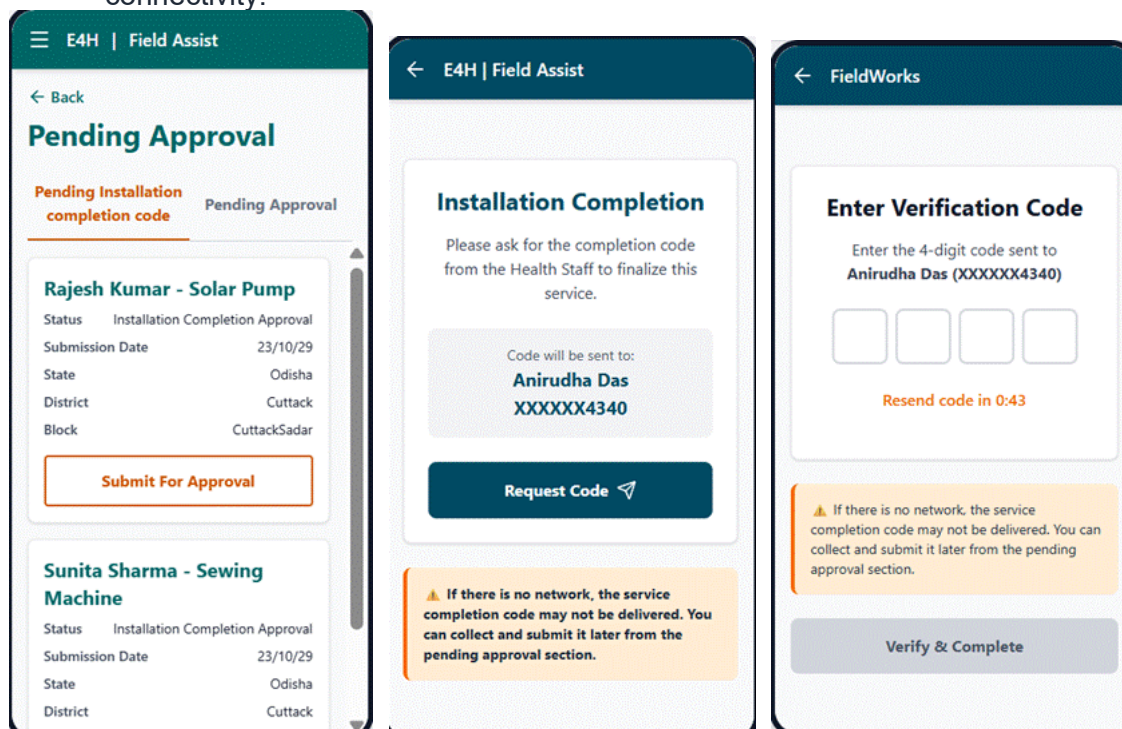
[UPDATED] Before an Installation Report can be finally submitted, the end user must confirm they received the training that came with the installation. This is done through an Installation Verification Code - a 4-digit one-time code sent by SMS to the end user's registered mobile number - so it cannot be skipped or done on the end user's behalf. Requesting this code needs network, since it's delivered by SMS - see below for what happens if none is available.

- [UPDATED] Once the report is filled in and ready, the technician taps Request Code. If there's network, the system sends the 4-digit Installation Verification Code by SMS to the end user's registered mobile number.
- The end user shares the code with the technician, who enters it and taps Verify & Complete.
- [UPDATED] If the code is verified, the report is submitted immediately into the Reviewer's queue (FR-12).
- [UPDATED] If there's no network to deliver the code, the completed report - all data, photos, and video already saved - simply waits under Pending Installation completion code (see Figure 8). Nothing is lost or has to be re-entered. The technician comes back to it once they're back online, taps Submit For Approval again to request the code, and completes submission from there.

Submission behaviour:

[UPDATED] Filling in and saving the report does not depend on the Installation Verification Code - a technician can complete and save a report with or without network, as described below. The code (above) is only needed as the final step that actually submits the report.

- Online path: if the technician has internet coverage when they finish, they request and enter the Installation Verification Code (above); once verified, the report is pushed directly to the backend and enters the assigned Reviewer's queue (FR-12) immediately.
- [UPDATED] Offline path: if there is no internet at the site, the technician can still fill in and save the report - including all photos and video - to local storage on the phone. The Installation Verification Code cannot be requested without network, so the report sits under Pending Installation completion code until the technician is back in a connected area, where they request and enter the code to complete submission.
- Sync is fully manual — the app does not attempt background auto-sync in Phase 1.
- The technician can also save a partially-filled draft and come back to it, regardless of connectivity.



[NEW] Figure 8. The actual Installation Verification Code flow. Left to right: the Pending Approval list, where Submit For Approval starts the request; the Request Code screen (needs network to send the SMS); and the 4-digit Enter Verification Code screen. If the code can't be delivered, the report stays under Pending Installation completion code until the technician retries with network.

7.5 Installation Review & Approval

FR-12: Review Queue

Installation Reviewers work through a queue of submitted IC Reports. The queue is scoped to the Plans they are assigned to, so a Reviewer never sees reports outside their responsibility. Each report opens with everything the Reviewer needs to make a decision — the technician's captured data, the photos and video, the template values as filled, and the generated Handover Letter — without needing to hunt across other systems.

Queue features:

- Shows only IC Reports for Installation Plans the Reviewer is assigned to.
- Each report displays all technician-captured data, photos/video, template values, and the Handover Letter.
- Filtering and sorting by Installation Plan, site, submission date, and status.

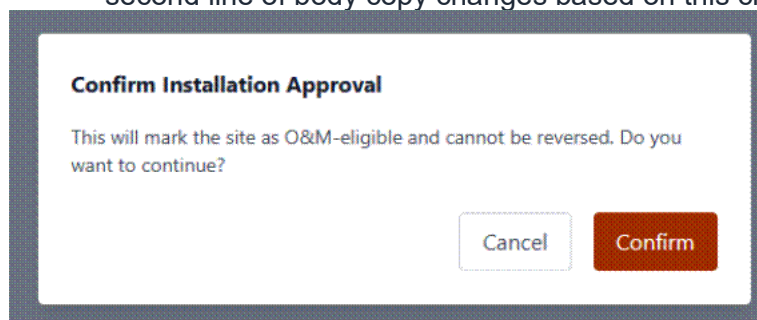
FR-13: Approval & Rejection

A Reviewer looks through an IC Report section by section — for example machine specs, photos, video, and Handover Letter. There is no separate Approve action per section; a section only needs the Reviewer's attention if something is wrong with it, in which case the Reviewer rejects that section and adds a reason. There is a single overall action button for the whole report. If the Reviewer has not rejected any section, that button reads Approve, and clicking it approves the report as a whole and fires downstream automation. As soon as the Reviewer rejects one or more sections, the button switches to Reject; clicking it sends the report back to the technician with the rejected sections and reasons attached, and they re-submit after fixing those sections.

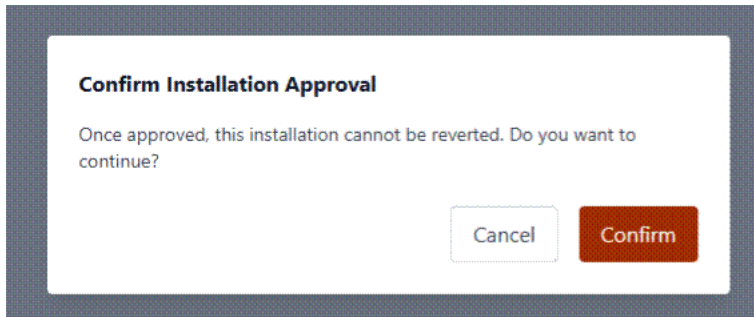
When a Reviewer clicks Approve (i.e. no section has been rejected), a confirmation modal appears before the action is finalized " This will mark it as Completed and cannot be undone. "The Reviewer must confirm before the approval is processed.

Modal copy depends on whether this is the first asset approved at this end user site:

- First asset approved at this site (no other asset at this site has been Approved before): the modal reads "Confirm Installation Approval" / "This will mark the site as O&M-eligible and cannot be reversed. Do you want to continue?" — flagging that this action is what first exposes the site on Setu4Livelihoods and the WhatsApp chatbot.
- Any subsequent asset approved at the same site (at least one other asset at this site was already Approved earlier): the modal reads "Confirm Installation Approval" / "Once approved, this installation cannot be reversed. Do you want to continue?" — a plain confirmation, since the site is already O&M-eligible and this step is only adding the second asset to it.
- Both variants are the same modal (title, buttons, and irreversibility warning) — only the second line of body copy changes based on this check.



First-asset-at-site variant — note the O&M-eligibility line.



Subsequent-asset variant — no O&M-eligibility line, since the site is already eligible.

Prototype link : [link](#)

Reviewer's actions:

- Per section: no action needed if the section is fine. Reject a section only if something is wrong with it, and add a reason.
- For the report overall: a single action button. It reads Approve while no section is rejected; it switches to Reject as soon as any section is rejected.

If the Reviewer clicks Reject (report is Returned):

- The list of rejected sections and reasons is captured on the report.
- The report goes back to the Field Technician for correction and re-submission.
- Technician re-submits; the report re-enters the Reviewer's queue.

If the Reviewer clicks Approve (no sections rejected):

- The installation is marked Completed.
- The end user site's overall status is re-evaluated. It moves to Completed only once every installation tied to that site has reached Completed status; otherwise, it remains In Progress.
- O&M eligibility is asset-level, not site-level: the specific asset just approved - a Machine or the Solar bundle - is immediately marked ready for O&M on Setu4Livelihoods, independently of any other assets at that site [UPDATED] (a site can have more than one Machine asset - see FR-01a). The end user can raise a maintenance ticket for it right away; no manual step is needed - the signal fires the moment that asset's IC Report is approved.
- [UPDATED] When another asset at the same end user site is approved later, it becomes eligible for O&M in the same way, independently. Over time this means the end user's list of raisable assets on Setu4Livelihoods grows one asset at a time, as each is approved — not all at once.
- This same asset list also drives the WhatsApp chatbot used for issue-raising: the moment an asset is approved, it becomes selectable in the chatbot's asset picker too, so the end user can raise a ticket for that asset through WhatsApp without any separate sync step.

[UPDATED] Worked example: a Machine + Solar site (most Solutions) becomes O&M-eligible in 2 independent steps - whichever asset is approved first becomes visible and raisable on Setu4Livelihoods immediately, the other only once its own IC Report is separately approved. A multi-machine site like Multi-Stage Processing (2 Machines + 1 Solar, see FR-01a) works the same way but in 3 independent steps instead of 2.

- The record flows to the Asset Management module (FR-14).

7.6 Post-installation

FR-14: Asset Management Handoff & Audit Trail

An approved installation does not end its lifecycle here — it flows into the Asset Management module for lifecycle tracking (maintenance, replacements, decommissioning). Additionally, every state-changing action across the full lifecycle is recorded in an audit trail, so any question about who did what and when can be answered without institutional memory.

Behaviours:

- On approval, the completed installation record becomes available in the Asset Management module.
- Every state-changing action — Project creation, Installation Plan creation, Vendor assignment, Template configuration, IC Report submission, Review actions — is recorded in the audit trail with actor, timestamp, and before/after values.
- The end user site lock (FR-06) releases once all installation reports for that site are Approved, making the site available for future Installation Plans.

7.7 Platform Support & Data Ingestion

FR-15: End User Site Data Ingestion

The Installation module does not maintain its own registry of end user sites. It draws from the Admin module of the broader platform, which is the single source of truth for site data, boundaries, and sunshine-hours mapping. This ingestion is a hard prerequisite — no Project can be created for a geography that has no ingested sites, because there would be nothing to include in the Project's scope.

What gets ingested:

- Site name, location (state/district/block), sector, value chain, programme details, and boundary information.
- Sunshine hours per site, derived from a state-wise mapping in the Admin module — when a state/district boundary is added there, it carries a sunshine hours value that each ingested site in that state inherits.
- Ingested sites become available for Project scope selection (FR-03).

FR-16: Platform Support

Different users need different platforms. Field Technicians work at rural installation sites on mobile devices, often offline, so they get a purpose-built Android app. Project Managers and Installation Reviewers work at desks with Excel files and browser-based dashboards, so they use the web app. A Field Technician web app is planned for Phase 2 to cover situations like tablets or shared workstations.

Phase 1 platform matrix:

- Android app — the Field Technician's primary tool. Supports full offline capture and manual sync (see FR-11).
- Web app — used by Project Managers and Installation Reviewers. Works on all screen sizes and supports all their tasks including the Excel upload/download flow.

- Field Technician web app is out of scope for Phase 1 (see Section 4.2).

9. Notification Matrix

[UPDATED] The app sends automated notifications across the Installation Plan and IC Report lifecycle (see matrix and templates below). All notifications listed in this section are sent via Email.

Trigger	Recipient	Channel
Installation Plan Published	Vendor Contact (Organisation Email)	Email
Installation Submitted for Review (IC Report Submitted)	Assigned Installation Reviewer	Email
IC Report Returned for Correction	Vendor Contact (Organisation Email)	Email + SMS
Installation Plan Entering Final Week (7 days before planned end date)	Program Manager	Email
Installation Plan Ending in 3 Days	Program Manager	Email
Installation Plan Overdue	Program Manager, Senior Programme Manager (out of platform role)	Email
Weekly Installation Progress Update	Program Manager	Weekly Email

[UPDATED] How these notifications work:

- Installation Submitted for Review and IC Report Returned are individual installation-level notifications and are triggered for each IC Report.
- Installation Plan reminders, escalation, and weekly updates are field plan-level notifications.
- 7-day reminder and 3-day reminder are sent only if the Installation Plan has pending installation activities.
- Overdue notification is sent only when the planned end date has passed and pending installation activities remain.
- Weekly Progress Update displays a single Installation Plan summary when only one plan is active; otherwise, it provides a consolidated view of all active Installation Plans.

[NEW] Notification Email Templates

[NEW] 1. Installation Plan Published

Subject: New Installation Tasks Available – {{Installation Plan ID}}

Body:

Hello,

A new Installation Plan has been published, and installation tasks have been assigned to your

organisation.

Project Name: {{Project Name}}
Installation Plan: {{Installation Plan ID}}
Total Installation Tasks: {{Total Installation Tasks}}
Planned Start Date: {{Start Date}}
Planned End Date: {{End Date}}

Please log in to the Setu4Livelihoods Installation App to view the assigned installation tasks and begin execution.

Regards,
Setu4Livelihoods

[NEW] 2. Installation Submitted for Review

Subject: Action Required: Installation Submitted for Review

Body:

Hello {{Reviewer Name}},

A new Installation Completion (IC) Report has been submitted and is awaiting your review.

Project Name: {{Project Name}}
Installation Plan: {{Installation Plan ID}}
End User: {{End User Name}}
Asset Type: {{Machine / Solar}}
Submitted By: {{Vendor Organisation}}
Submission Date: {{Submission Date & Time}}

Please review the submitted IC Report.

Regards,
Setu4Livelihoods

[NEW] 3. IC Report Returned for Correction

Subject: IC Report Returned for Correction – {{End User Name}}

Body:

Hello,

The submitted Installation Completion (IC) Report has been returned for corrections.

Project Name: {{Project Name}}
Installation Plan: {{Installation Plan ID}}
End User: {{End User Name}}
Asset Type: {{Machine / Solar}}

Please log in to the Setu4Livelihoods Installation App to review the rejection comments, make the required corrections, and resubmit the IC Report.

Regards,
Setu4Livelihoods

[NEW] 4. Installation Plan Entering Final Week

Subject: Reminder: Installation Plan Entering Final Week

Body:

Hello {{Program Manager}},

Your Installation Plan is entering its final week and still has pending installation activities.

Project Name: {{Project Name}}

Installation Plan: {{Installation Plan ID}}

Current Progress

Total Installation Tasks: {{Total}}

Completed: {{Completed}}

Pending Review: {{Pending Review}}

Pending Installation: {{Pending Installation}}

Completion: {{Completion %}}

Planned End Date: {{End Date}}

Please review the current progress and plan the remaining installation activities to ensure completion by the planned end date.

Regards,

Setu4Livelihoods

[NEW] 5. Installation Plan Ending in 3 Days

Subject: Reminder: Installation Plan Ending in 3 Days

Body:

Hello {{Program Manager}},

Your Installation Plan is approaching its planned end date and still has pending installation activities.

Project Name: {{Project Name}}

Installation Plan: {{Installation Plan ID}}

Current Progress

Total Installation Tasks: {{Total}}

Completed: {{Completed}}

Pending Review: {{Pending Review}}

Pending Installation: {{Pending Installation}}

Completion: {{Completion %}}

Planned End Date: {{End Date}}

Immediate attention is required to complete the remaining installation activities before the planned end date.

Regards,

Setu4Livelihoods

[NEW] 6. Installation Plan Overdue

Subject: Escalation: Installation Plan Overdue

Body:

Hello,

The following Installation Plan has exceeded its planned completion date and still has pending installation activities.

Project Name: {{Project Name}}

Installation Plan: {{Installation Plan ID}}

Planned End Date: {{End Date}}

Days Overdue: {{Days Overdue}}

Current Progress

Total Installation Tasks: {{Total}}

Completed: {{Completed}}

Pending Review: {{Pending Review}}

Pending Installation: {{Pending Installation}}

Completion: {{Completion %}}

Please review the delay and initiate the necessary corrective actions to complete the remaining installation activities.

Regards,

Setu4Livelihoods

[NEW] 7. Weekly Installation Progress Update

Scenario 1: Single Active Installation Plan

Subject: Weekly Installation Progress Update

Body:

Hello {{Program Manager}},

Below is the current progress of your Installation Plan.

Project Name: {{Project Name}}

Installation Plan: {{Installation Plan ID}}

Metric	Value
Total Installation Tasks	{{Total}}
Completed	{{Completed}}
Pending Review	{{Pending Review}}
Pending Installation	{{Pending Installation}}
Completion	{{Completion %}}
Planned End Date	{{End Date}}
Status	{{On Track / Needs Attention / Overdue / Completed}}

Please review the current progress and plan the remaining installation activities accordingly.

Regards,

Setu4Livelihoods

Scenario 2: Multiple Active Installation Plans

Subject: Weekly Installation Progress Update

Body:

Hello {{Program Manager}},

Below is the current progress of your active Installation Plans.

Installation Plan	Project	Completion	Pending Review	Pending Installation	End Date	Status
{{IP-001}}	{{Project A}}	{{82%}}	{{2}}	{{8}}	{{12 Aug}}	On Track
{{IP-002}}	{{Project B}}	{{45%}}	{{5}}	{{22}}	{{08 Aug}}	Needs Attention
{{IP-003}}	{{Project C}}	{{100%}}	{{0}}	{{0}}	Completed	Completed

Overall Summary

Active Installation Plans: {{Count}}

Total Installation Tasks Completed: {{Completed}}

Total Pending Installation Tasks: {{Pending}}

Installation Plans Entering Final Week: {{Final Week}}

Installation Plans Ending in 3 Days: {{Ending Soon}}

Overdue Installation Plans: {{Overdue}}

Please review the current progress and prioritize Installation Plans requiring attention.

Regards,

Setu4Livelihoods

10. Workflows

10.1 Project & Installation Plan Preparation (Project Manager)

Step	Who	What happens
1	Project Manager	Creates a Project: enters name, justification, dates, geography. System reserves a Project ID.
2	Project Manager	Downloads the End User Site Excel, marks Include (Yes/No), uploads. System finalises Project scope and generates the Project ID.
3	Project Manager	Creates an Installation Plan under the Project: enters plan geography, sector(s), Reviewer, dates, and Senior Programme Contact.
4	Project Manager	Downloads the Installation Scope Excel, marks Include (Yes/No), picks a Solution per included row (dropdowns filtered per FR-01), uploads. System validates lock (FR-06) and Solution rules; on success, generates the Vendor Assignment Excel and the Installation Template list.

5	Project Manager	Downloads the Vendor Assignment Excel, assigns Vendor and Vendor Email per row, uploads.
6	Project Manager	Downloads one blank template per unique Solution, fills machine and solar sections, uploads each.
7	Project Manager	Runs Publish validation, submits. System publishes the Plan and dispatches tasks to assigned Vendors.

10.2 Installation Execution & IC Report Submission (Field Technician)

Step	Who	What happens
1	Field Technician	Opens the assigned task in the Android app.
2	System	Pre-fills site details, asset specs, and template data prepared by the PM. Technician sees only the section relevant to their assignment.
3	Field Technician	Completes the installation on-site. Can work fully offline — fill fields, upload photos/video, save locally.
4	Field Technician	Fills the IC Report (auto-selected format), confirms/edits pre-filled values, reviews the generated Handover Letter.
5	Field Technician	Submits the IC Report if online, or saves it locally to sync later.
6	Field Technician	When back in an internet-accessible area, taps Sync to push saved reports.
7	System	Sends the report to the assigned Reviewer's queue and triggers notifications per Section 9.

10.3 Review, Approval & Post-Installation (Installation Reviewer)

Step	Who	What happens
1	Reviewer	Opens the IC Report from the queue and checks all data, photos, and the Handover Letter.
2a	Reviewer	If something is wrong: Returns the report with a reason. Field Technician is notified to fix and re-submit.
2b	Reviewer	If everything is correct: Approves the report.
3	System	Installation status is set to Completed.
5	System	[UPDATED] "The asset just approved (a Machine, or Solar) is immediately marked as ready for O&M on Setu4Livelihoods and in the WhatsApp chatbot's asset picker, independent of any other assets at the site (a site can have more than one Machine asset - see FR-01a). The end user can raise a maintenance ticket for it right

		away; each remaining asset becomes raisable the same way, independently, once it, too, is approved."
6	System	Completed record flows to the Asset Management module (FR-14); audit trail is finalised.
7	System	End user site lock (FR-06) releases if all reports for that site are now Approved.

11. Glossary

Term	Definition
IC Report	Installation Completion Report — the structured form filled by a Field Technician to document a completed installation.
Installation Plan	An execution plan created under a Project that covers one or more end user sites and dispatches installation tasks to assigned Vendors.
End User Site	The physical installation location associated with an end user; the unit that Projects and Installation Plans operate on.
Solution	[UPDATED] A pre-configured package of one or more machines plus an associated solar bundle, designed for a specific livelihood use case (e.g. Pulverizer, Grinder). Most Solutions have one machine; some, like Multi-Stage Processing, have more than one - each machine becomes its own separate installation (see FR-01a).
Sheet 1 / Installation Scope Excel	The Excel used in FR-05: pick which sites are in the Installation Plan and assign a Solution to each.
Sheet 2 / Vendor Assignment Excel	The Excel used in FR-07: assign a Vendor Organisation and Vendor Email to each expanded asset row (Machine + Solar per site).
Installation Template	The data ingestion form the PM fills per unique Solution; pre-populates IC Report fields for the Field Technician.
Setu4Livelihoods	The Livelihood Issue Resolution Platform; the companion platform for post-installation O&M support.
Handover Letter	Formal document handed to the end user at the end of a successful installation.
Admin Module	The platform module that maintains state/district/block boundaries, sunshine-hours mapping, and user/role configuration.
Asset Management Module	The downstream module where approved installation records live for lifecycle tracking (FR-14).
DRE	Distributed Renewable Energy — the technology domain underlying SELCO's livelihood programme.
[NEW] Image Capture Master	A reference list, maintained separately, mapping each required installation photo to the Solution component it belongs to - this is what lets the Installation Report's photo checklist (FR-11b) show only

	the photos relevant to the selected Solution, plus the two always-mandatory photos.
--	---

12. Appendix: User Workflows

The three diagrams below show the end-to-end journey for each of the three user roles in the app. They are meant to anchor engineering walkthroughs — one page per stakeholder, showing every step from start to finish.

12.1 Project Manager

The Project Manager's workflow covers eight steps across three phases: Project Setup, Plan Preparation, and Publish. Each step is a discrete action; the sequence is enforced (Publish only fires after all preceding steps are complete).

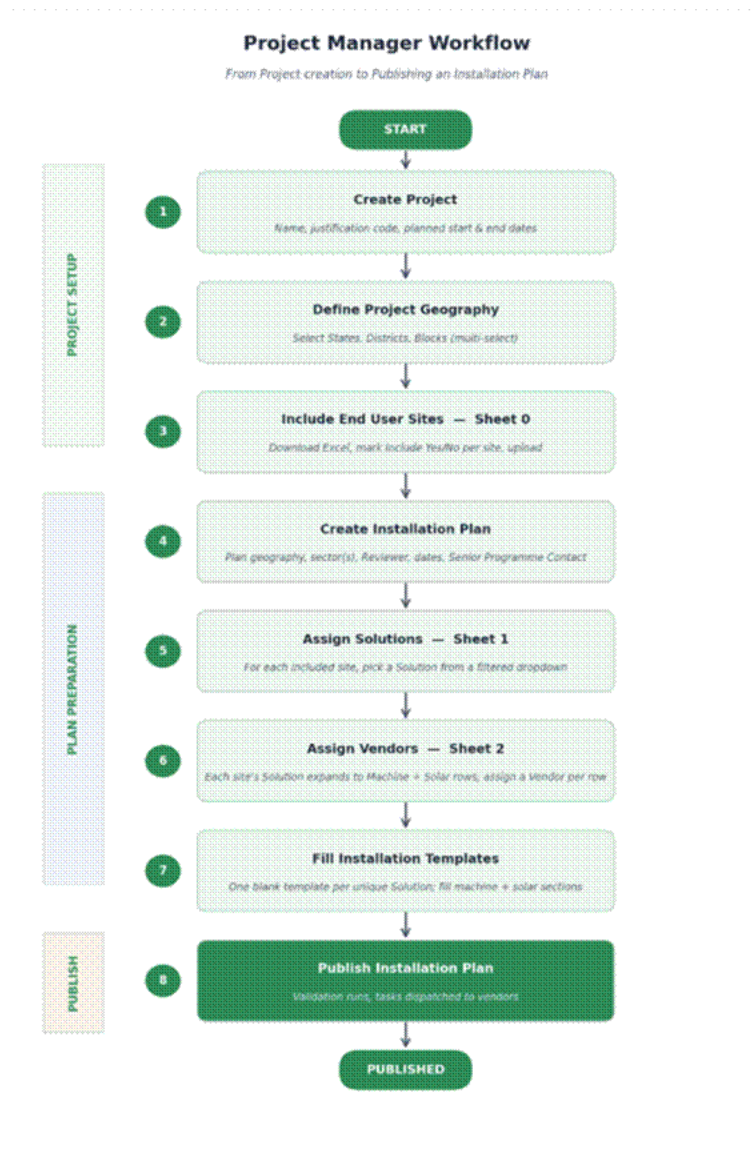


Figure 3. Project Manager workflow.

12.2 Field Technician

The Field Technician's workflow spans three phases: Task Pickup, On-Site installation (offline-capable), and Submission. After filling the IC Report, the technician takes one of two paths depending on internet availability — direct submission when online, or Save Locally + Sync when offline. Both paths converge at the report reaching the Reviewer's queue.

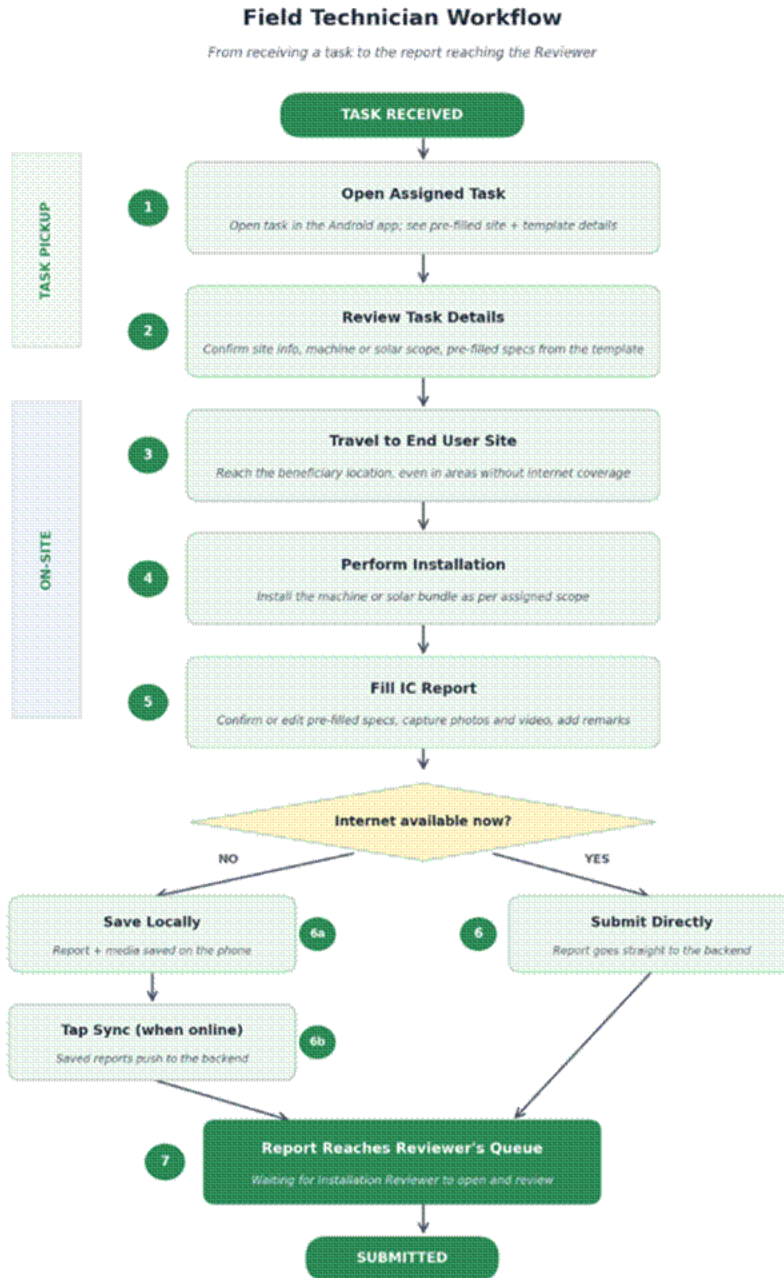


Figure 4. Field Technician workflow.

12.3 Installation Reviewer

The Installation Reviewer opens each IC Report and reviews it section by section, rejecting (with a reason) any section that's wrong. If one or more sections are rejected, the report goes back to the technician for correction and re-submission — it re-enters the review queue as a fresh submission. If no sections are rejected, the Reviewer approves the report as a whole and downstream actions fire automatically.

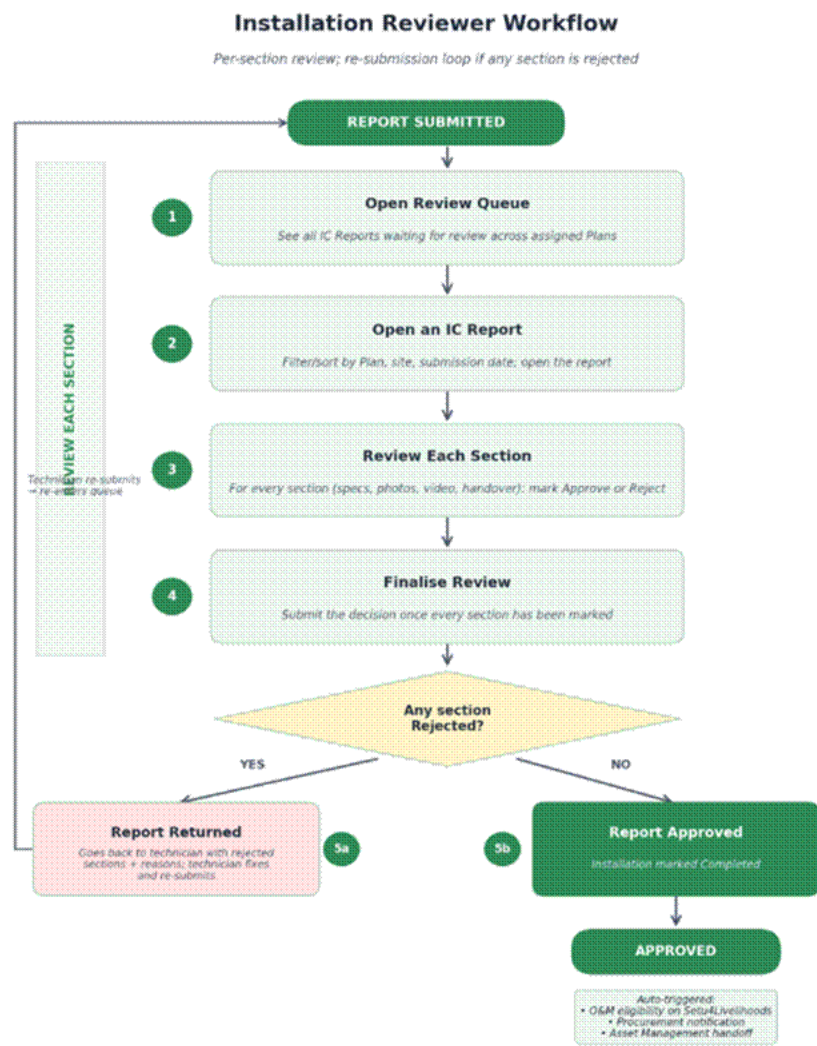


Figure 5. Installation Reviewer workflow.

[UPDATED] 13. Appendix: The Solution Ingestion Sheet (Parts List by Solution)

[UPDATED] Despite its filename ("ICC Report Format by Solution"), this workbook is the Solution Ingestion Sheet, not the IC Report's actual layout - it just supplies the parts list and installation details for each Solution (feeds FR-01a, FR-08). The real IC Report has its own separate format (FR-11, Section 13.5). Section 13.4 lists the 14 Solutions this sheet currently covers.

[UPDATED] 13.1 What's in the Ingestion Sheet

[UPDATED] The ingestion sheet - not the IC Report - is organised into the four parts below; it supplies the data an IC Report draws on (FR-11a), not the IC Report's own layout (FR-11).

- Site & Document Details: basic information like the Tender/Purchase Order number, Report number, Date, the installation site's name and address (Block/Taluk, District, State, Pincode), its GPS location, Invoice number, Solution name, and Bundle/Item Code.
- Solar Parts List: a simple table (Number / Product / Make / Size / Quantity) grouped by category (see Section 13.2). Only the categories and products that apply to the chosen Solution show up.
- Associated Machines: the same kind of table, listing the livelihood machine(s) that come with the Solution (for example, a Pulverizer, Sewing Machine, or Rice Huller) - these two sections are also what the system reads to auto-sort Machine parts from Solar parts (see FR-01a).
- System Functionality Parameters: the readings the Field Technician takes on-site when setting up the system (see Section 13.3).

[NEW] 13.2 Solar Parts List - Categories

The Solar parts are grouped into the categories below. Not every category applies to every Solution - a category only shows up in the report if that Solution actually uses it (FR-11a, point 5).

- Solar Panel
- Battery
- Inverter / PCU
- Mounting Structure
- Rack / Enclosure
- Junction / Protection Box
- Cable
- Switch / Socket / MCB
- Lighting / Fan
- Lightning Protection (applicable to select Solutions only)
- Earthing
- Safety / Docs
- Fire Extinguisher (applicable to select Solutions only)
- Consumables

[NEW] 13.3 System Functionality Parameters

Taken once per installation, no matter which Solution it is:

- Date and Time of recording; Weather Condition at the time of recording.
- Array Details: Array Size (kWp), No. of modules, No. of strings, No. of modules per string, Each module capacity (Wp).
- Array Parameters (at the AJB): Voc, Vmp, Imp, Pmp per string and at array level, under specified Grid/MCB test conditions.
- Battery Bank Parameters (Grid OFF, Load ON): Battery Bank Voltage and Current.

- PCU / Inverter Parameters (1-Phase / 3-Phase as applicable): load/output voltage, output frequency, output current at full load — measured with Grid OFF and Grid ON.
- PCU / Inverter Display & Setting: Inverter priority setting, Grid export feature, Battery settings (Ah / type), and what the load is currently running on.
- Changeover Switch Settings: switch orientation and functional check for both Solar and Grid.
- Measurements at Sockets: voltage between Phase-Neutral, Phase-Earth, and Earth-Neutral.
- Earth Pit Tests (standalone, before connecting to system): resistance readings for DC Earthing, AC Earthing, and each Lightning Arrester.

[NEW] 13.4 Solutions Covered in Phase 1

[UPDATED] The table lists the 14 Solutions we're starting with (FR-01), each with its ingestion sheet ready and Bundle/Item Code.

#	Solution	Bundle / Item Code
1	Textile Lighting	202526PASF0000104
2	Pulverizer	202526PASF0000379
3	Printer	202526PASF0000540
4	Sewing Machine	202526PASF0000135
5	LSK (Laptop)	202526PASF0000214
6	Eri Spinning	202526PASF0000141
7	Paddy Integrated Processing	202526PASF0000382
8	Multi-Stage Processing (Millet)	202526PASF0000395
9	Refrigerator	202526PASF0000248
10	Multi-Stage Processing	202526PASF0000387
11	Light Manufacturing	202526PASF0000460
12	Roaster	202526PASF0000317
13	Rice huller	202526PASF0000371
14	Oil Mill	202526PASF0000390

[UPDATED] New Solutions get added the same way (FR-01) - no PRD changes needed.

[NEW] 13.5 Who Fills In What, and What the System Fills In Automatically

Taken from the “Templates for Installation App - Livelihood” reference document. This explains who is allowed to fill in or edit each part of the template and IC Report, and which parts the system fills in automatically. The source document describes this for the Solar template/report; we're assuming the same rules apply to the Machine section too, unless the source document says otherwise.

Field / Section	Who populates or edits it	When
Bill of Materials (BOM)	Project Manager (initial fill); Field Technician (edits/enters during installation)	Template configuration (FR-08) and IC Report filling (FR-11)
System Parameters	Field Technician only	During installation (on-site readings)
Tender Number	Program Team / Project Manager	While downloading / uploading the solution ingestion template
PO Number / WO Number	Field Technician	During installation
Report Number	System (auto-generated)	On IC Report creation
Invoice Number	Field Technician	During installation
Capacity & Quantity (per component)	System (pre-populated based on selected Solution); Project Manager can modify by adding/removing quantities	Template download / configuration (FR-08)
Annexure Section	System (auto-populated from serial numbers scanned/entered by the Field Technician)	During installation
Site Details — Name, Block/Taluk, District, State, Pincode	System (auto-populated from End User Ingestion and Project Creation data)	Automatic, on report generation

[UPDATED] Clarified: the Annexure (the serial-number table for Battery, Inverter, and Panels) is not part of the Solution Ingestion Sheet - it only exists in the final Solar Installation Completion Report. The Field Technician scans or enters the serial numbers on-site, and the Annexure is built from that data once the technician has reviewed the pre-filled details, made any corrections, and submitted the report. The vendor sees the Annexure in that final, submitted report - not in the ingestion sheet.

Reference documents

Linked from the source document - these are placeholders until the programme team shares the real links:

- Ingestion templates per Solution
- [UPDATED] Solar IC Report: [Solar IC report.docx](#)
- [UPDATED] Machine IC Report: [Machine Installation Report.docx](#)

[NEW] 13.6 Solar IC Report - Actual Report Structure

Full document: [Solar IC report.docx](#). Summary of what the real, final Solar Installation Completion Report contains - this is what the Field Technician actually fills in and submits, not the Solution Ingestion Sheet described in 13.1-13.4:

- Site & document details: Purchase/Work Order No., Tender no., Report No., Date, site name and address, Geo-Coordinates, Invoice number.
- Bill of Material (Solar System): the solar parts list (Sl. No. / Product / Make / Capacity / Quantity).
- Bill of Material (Luminaries & Fans): LED bulbs, tube lights, ceiling and wall fans, outdoor lights.

- Bill of Material (RMS): only included if a Remote Monitoring System is installed - covers the RMS model, type, IP/MAC ID, serial number, SIM number, and installation/network status.
- Bill of Material (Load Wiring): the AC-side wiring items - distribution box, RCCBs, isolators, MCBs, fan regulators, sockets, switches, and power cables.
- System Functionality Parameters: the same on-site readings described in Section 13.3 (Array Details, Array/Battery/PCU parameters, socket measurements, earth pit tests), plus separate tables for 1-Phase and 3-Phase systems.
- Orientation Checklist: whether the end user was shown basic operation and routine maintenance, and whether the safety posters (No Fire, PASS, Danger, Do's & Don'ts with helpline number) were put up.
- [UPDATED] Required photos: generated dynamically for the Solution installed, via the Image Capture Master (see FR-11b, and the reference table in Section 13.8) - only the photo entries whose component is present in that Solution are shown (e.g. solar panels with mounting structure, batteries, inverter, cable routing, AJB, GIPB), each with the number of photos required. The end-user-place and end-user-with-Solar-System photos are always required, for every Solution.
- End user and Technician signatures.
- Annexure - Serial Numbers: separate serial-number tables for the Solar Panel, Solar Battery, and Solar Inverter, each with Make and Capacity per unit. This is the Annexure discussed in Section 13.5 - it only exists in this final report, not in the Solution Ingestion Sheet.

[NEW] 13.7 Machine IC Report - Actual Report Structure

Full document: [Machine Installation Report.docx](#). Much shorter than the Solar IC Report - it has no Bill of Materials table, no system-parameter readings, and no Annexure:

- Site & document details: Location, Date, PO Number, Machine Serial Number, Manufacturer Invoice Number, Machine Specifications/Motor Capacity, Warranty (years).
- Installation Verification: Civil Work (2 photos), Electric/Switch Board (1 photo), Demo Test with Raw Material (1 video) - each item just needs the required photo/video attached.
- End User Handover: a photo with the end user, and whether the end user was trained (Yes/No).
- End user and Technician signatures.

Note: because the Machine report is this much simpler, the Machine installation task in the Field Assist app (FR-10, FR-11) needs far fewer fields than the Solar task for the same end user site.

[NEW] 13.8 Image Capture Master - Reference Table

This is the actual Image Capture Master behind FR-11b and the "Required photos" line in Section 13.6 - it maps each Solar component to the photo(s) required for it. Only the rows whose component is present in the selected Solution are shown in that Solution's IC Report (FR-01a, FR-11b); the last two rows (Mandatory) always show, matching FR-11b's rule 6.

Sl. No.	Main Heading	Image Details (for image upload)	Required No. of Images	Tick if taken	Remarks
1	Solar Panel	Clear image of solar panels with Module mounting structure from a range which gives better visibility (capture with standard marking).	2		

2	Battery	Clear image of batteries from a range which gives better visibility, including the water level (capture with standard marking).	1		
3	Inverter	Clear image of inverter from a range which gives better visibility (front and back) (capture with standard marking).	2		
4	Mounting Structure	Clear image of the inverter isolator switches & Load MCB.	1		
5	Rack/Enclosure	Clear image of cable routing for the entire system (capture with standard marking).	3		
6	Junction/Protection Box	Clear image of AJB.	1		
8	Cables	Clear image of Lightning Arrestor.	1		
9	Switch/Socket/MCB	Clear image of Earthing pits.	1		
10	Earthing	Clear image of Changeover Switch.	1		
11	Safety Docs	Clear image of DO's and Don'ts Poster.	1		
12	Lighting/Fan	Clear image of Foam Plaques (SLD, High Voltage, PASS, No Fire, Danger, Risk of Electric Shock).	2		
15	RMS	RMS site photo with geotagging.	1		
16	Mandatory	Clear image of the End User place (Long Shot).	1		
17	Mandatory	Clear image of End User with Solar System.	1		